Experiment- 10

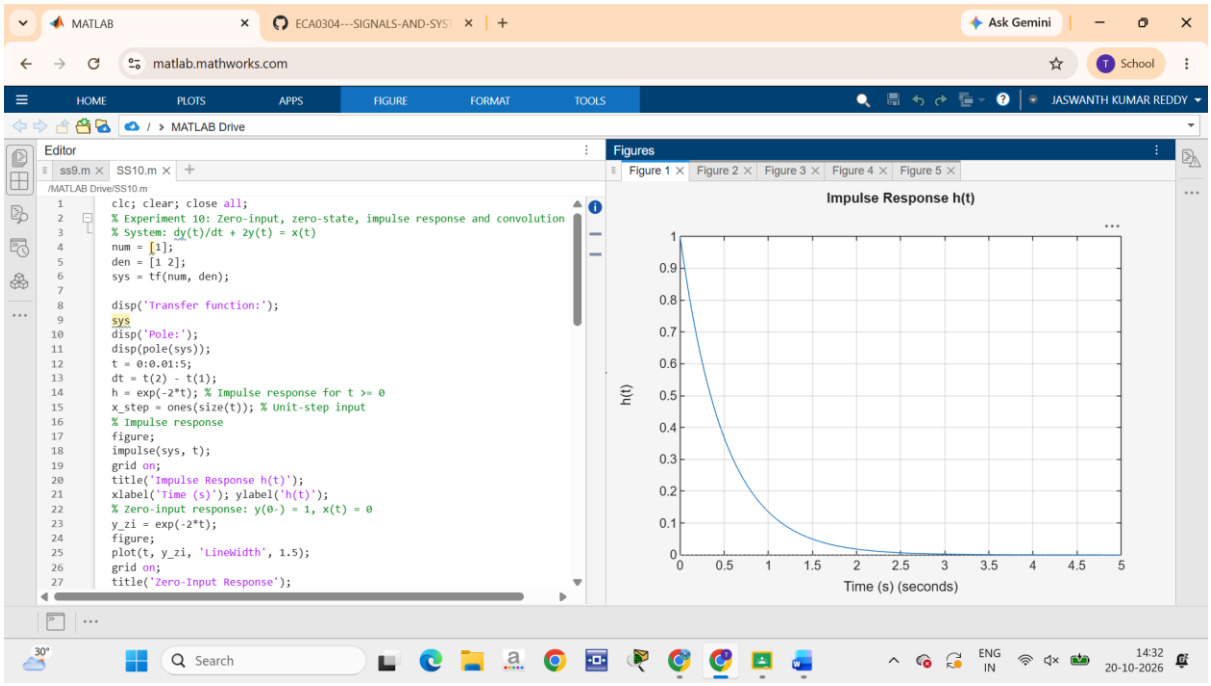
Zero-State and Zero-Input Response, Impulse Response and Convolution

OBJECTIVE

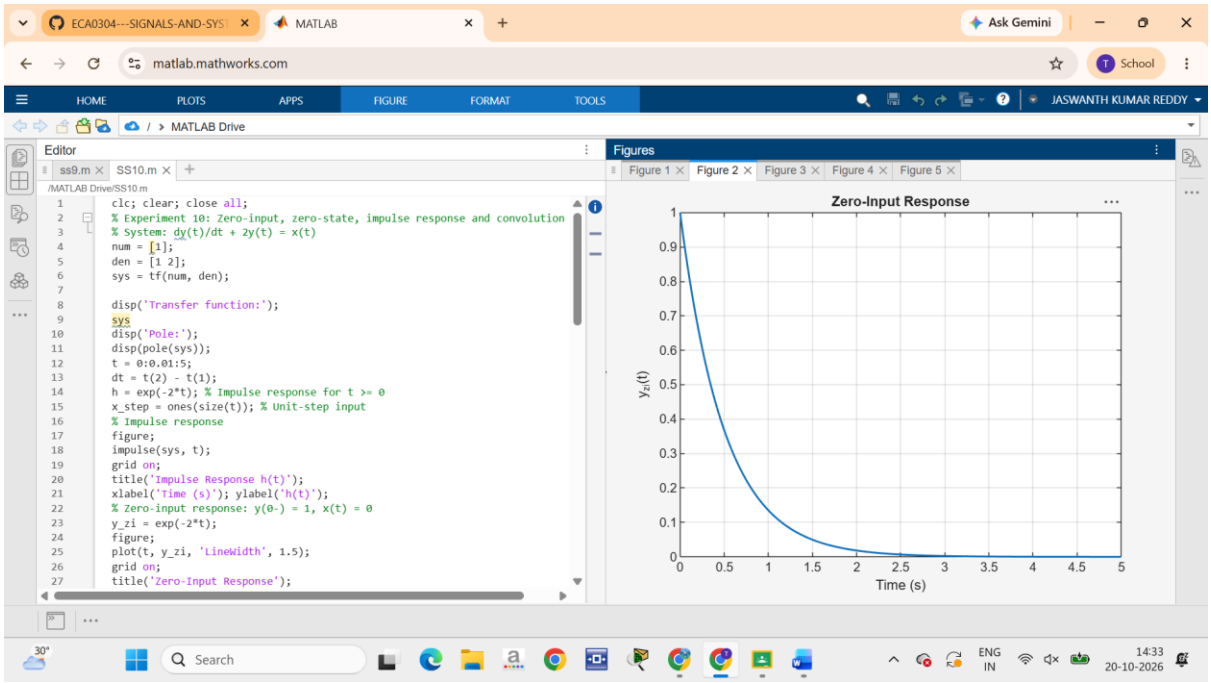
• To distinguish the responses produced by initial stored energy and external excitation, obtain the impulse response of a continuous-time LTI system, and verify that the total response is the sum of the zero-input and zero-state responses, with the zero-state response also obtained through convolution.

Output graphs:

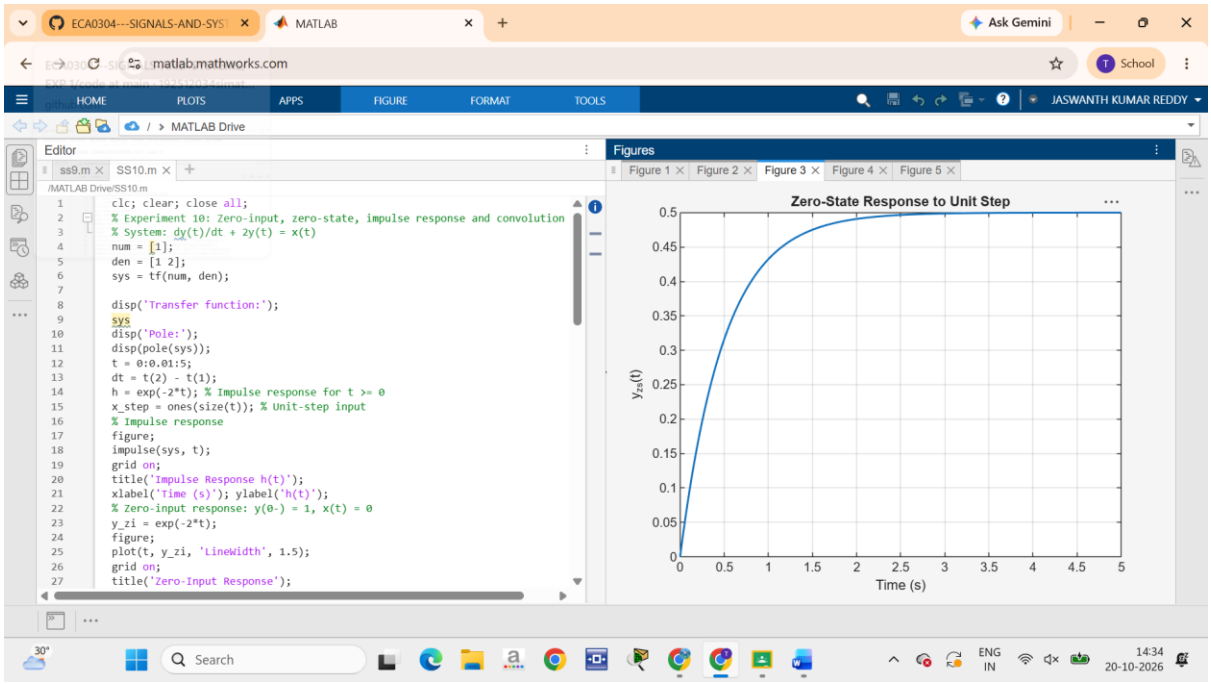
Impulse response:



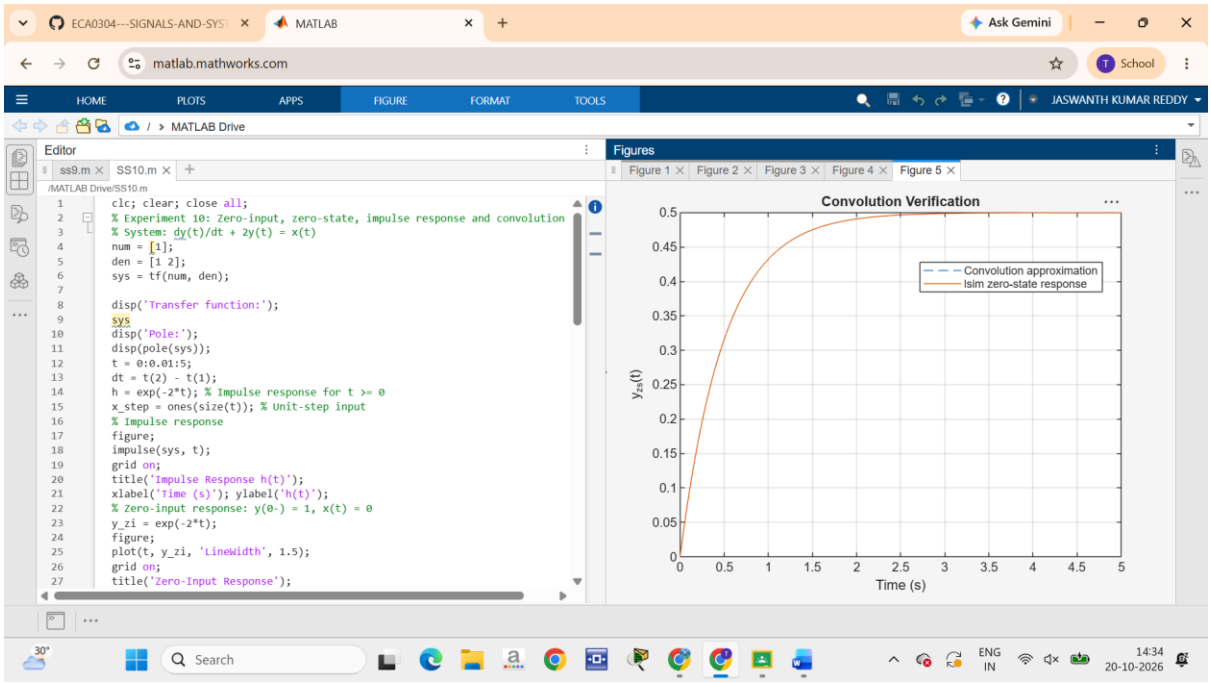
Zero-input :

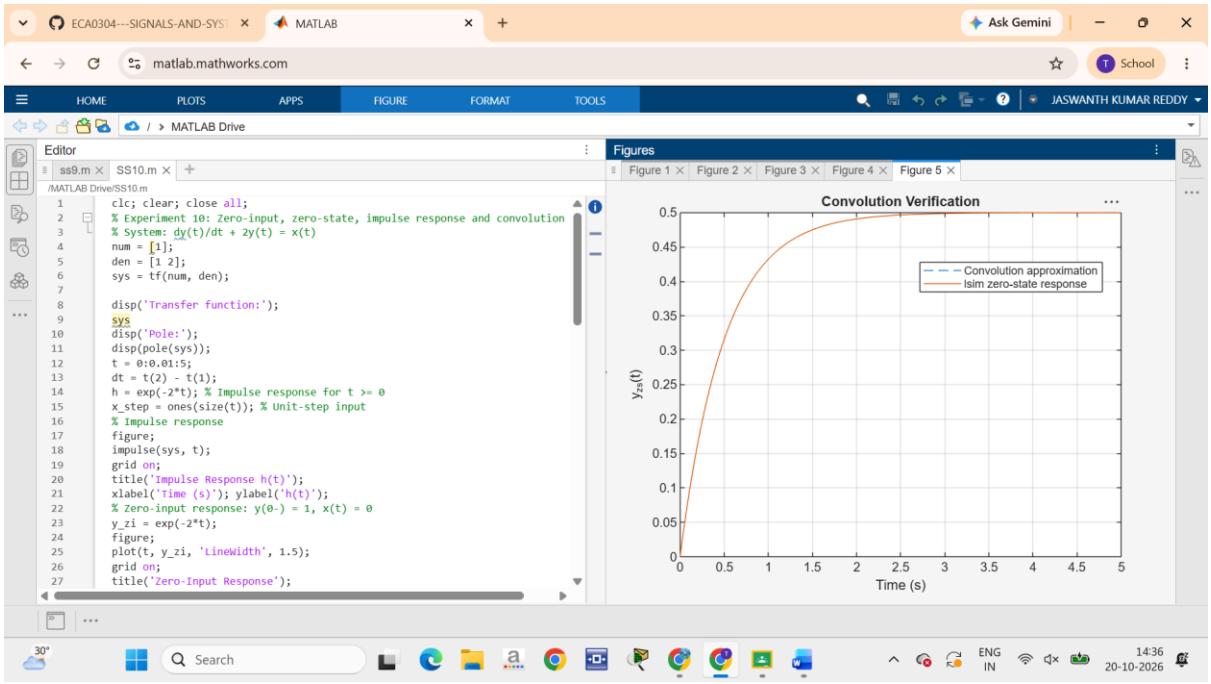


zero-state response:



Response-component comparison:





Result: Zero-state response, zero-input response, impulse response, and system output using convolution were analyzed for continuous-time LTI systems.